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CHENNAI

Reg No 23BCE1034

Final Assessment Test(FAT) - APR/MAY 2026

Programme	**B.Tech.**	Semester	**Winter Semester 2025-26**
Course Code	**BCSE303L**	Faculty Name	**Prof. Karthik R**
Course Title	**Operating Systems**	Slot	**D1+TD1**
		Class Nbr	**CH2025260502127**
Time	**3 hours**	Max. Marks	**100**

Instructions To Candidates

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

Course Outcomes

CO1: Interpret the evolution of OS functionality, structures, layers and apply various types of system calls of various process states.
CO2: Design scheduling algorithms to compute and compare various scheduling criteria.
CO3: Apply and analyze communication between inter process and synchronization techniques.
CO4: Implement page replacement algorithms, memory management problems and segmentation.
CO5: Differentiate the file systems for applying different allocation, access technique, representing virtualization and providing protection and security to OS.

Section - I
Answer all Questions (4 × 10 Marks)

01. An intern has recently joined a software company and is assigned to understand how a computer system functions at the system level. As part of the onboarding process, the intern is asked to study how the system supports interaction through different interfaces, enables applications to run, and manages device-related operations while ensuring smooth and stable performance. Based on this scenario, explain any five different services provided by an Operating System

[10] (CO1/K1)

02. A storage system is being evaluated to understand how files are organized on disk. It is observed that some files occupy adjacent storage locations enabling fast access, while others are distributed across different locations but remain logically connected. In another case, an additional structure is used to keep track of all the locations associated with a file. These differences influence how efficiently files are accessed, stored, and managed. Based on this scenario, explain the different file allocation methods used by an Operating System.

[10] (CO5/K2)

03. A system-level utility is being developed in a Unix-based environment to perform two independent tasks simultaneously. One task involves generating and displaying all even numbers up to a given limit with a fixed time delay between each output, while the other task involves displaying each character of a given name with a larger delay between successive outputs. These tasks must be executed in separate execution paths, and the main program should coordinate their execution such that it waits until both tasks are completed before displaying a final confirmation message. Based on this scenario, write a Process-based C program to implement the required functionality.

[10] (CO3/K3)

04. A smart warehouse system uses automated robots to manage package handling. Multiple Loader robots continuously place packages into a shared conveyor buffer, while multiple Dispatcher robots pick packages from the buffer for delivery. The conveyor belt has a fixed capacity of N packages, and due to high concurrency, issues like buffer overflow and underflow have been observed. To solve this, the system architect proposes a monitor-based synchronization mechanism. As an operating systems expert, you are required to design a monitor construct for this system by clearly specifying the shared variables, condition variables, and write appropriate pseudocode for the Producer and Consumer operations.
Further, you must explain how your monitor solution ensures mutual exclusion and prevents buffer overflow and underflow conditions, and justify how condition variables are used effectively to coordinate between producers and consumers, ensuring that processes wait and resume correctly without busy waiting.

[10] (CO3/K3)

Section - II

Answer all Questions (4 × 15 Marks)

05. A high-performance computing team at DEF University is analyzing the memory behavior of a real-time satellite image processing system. Due to hardware constraints, the system currently operates with 4 page frames in main memory. During a peak execution phase, the processor generates the following page reference string:

1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5, 6, 1, 2, 3, 4, 5, 6, 7, 1, 2, 3

To evaluate system performance under strict latency requirements, you are required to apply FIFO, LRU, and Optimal page replacement algorithms for 4 frames, clearly showing all page replacement decisions and computing the total number of page faults for each algorithm.

Further, the system is proposed to be upgraded to 5 page frames. You must apply the FIFO algorithm again for 5 frames using the same reference string and determine whether Belady's Anomaly occurs by comparing the page fault counts for 4 and 5 frames.

[15] (CO4/K3)

06. A cloud-based database server supporting an e-commerce platform uses a disk system with 250 cylinders (0–249). During a flash sale, the disk controller receives a burst of I/O requests arriving dynamically in the following order:

(120, 15, 180, 90, 30, 200, 75, 220, 10, 160, 85, 140, 60)

The initial head position is at cylinder 100, and the disk was previously moving towards higher-numbered cylinders. Due to strict performance requirements, the system administrator wants to evaluate different scheduling strategies.

Apply FCFS, SSTF, SCAN, C-SCAN, LOOK, and C-LOOK disk scheduling algorithms for the above-mentioned request list and compute the total seek distance and show the complete sequence of head movements. Which algorithm minimizes seek time and state which algorithm provides better fairness for distant requests ? Finally, briefly explain why SSTF may not always be suitable in real-time systems, even if it gives a lower seek time in some cases.

[15] (CO5/K3)

07. A software company operates a Continuous Integration / Continuous Delivery (CI/CD) build server where multiple automation jobs run concurrently. Each job must acquire exclusive locks 'L' on shared tools before proceeding:

L1 = Build Cache Lock
L2 = Package Repository Lock
L3 = Test Database Lock
L4 = Deployment Token Lock

At a particular time, the system snapshot is:

Current Holdings (who holds what)
P0 holds L1
P1 holds L2
P2 holds L3
P3 holds L4

Current Requests (who is waiting for what)
P0 requests L2
P1 requests L3
P2 requests L1
P3 requests L2

To decide an appropriate recovery action, the system also records the following:

Process	Priority (1 = highest)	Rollback Cost	Progress Completed	Resources Held
P0	2	Medium	70%	1
P1	1	High	20%	1
P2	3	Low	80%	1
P3	2	Low	10%	1

(a) Draw the Resource Allocation Graph (RAG) for the given snapshot. [4 marks]
(b) Convert the RAG into a Wait-For Graph (WFG) and list all WFG edges. [4 marks]
(c) Determine whether a deadlock exists. If yes, write:
- the cycle in the WFG, and

- the set of deadlocked processes. [3 marks]

(d) The admin wants to recover with minimum impact. Choose one recovery action and justify your choice using at least two factors from the table:

- Abort/Rollback one process (state which), OR
- Preempt one lock (state which lock and from whom). [4 marks]

[15] (CO3/K3)

08. A customer-support ticketing platform runs multiple backend jobs. Each job has an Arrival Time (AT), CPU Burst Time (BT), and a Priority (PR). Lower priority number means higher priority (i.e., PR = 1 is the highest). Assume CPU switching time = 0 and all bursts are CPU bursts only (no I/O).

Process Table

Process	Arrival Time (ms)	Burst Time (ms)	Priority (PR)
P1	0	6	3
P2	1	4	1
P3	2	5	4
P4	3	2	2
P5	4	3	1
P6	6	4	2

Tie-breaking rule:

If two processes have the same priority, schedule the one with earlier arrival time; if still tied, use FCFS.

Using the given process table, solve the CPU scheduling problem using Priority Scheduling in both modes.

i) Apply Non-Preemptive Priority Scheduling: draw the Gantt chart, compute for each process the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT), and then calculate the Average Waiting Time and Average Turnaround Time. [7 marks]

ii) Apply Preemptive Priority Scheduling for the same dataset: draw the Gantt chart clearly indicating all preemptions, compute CT, TAT, and WT for every process, and calculate the Average Waiting Time and Average Turnaround Time. [7 marks]

iii) Based on your computed averages, compare both approaches and state which one results in a lower average waiting time for this dataset, with proper justification. [1 mark]

[15] (CO2/K3)

BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)